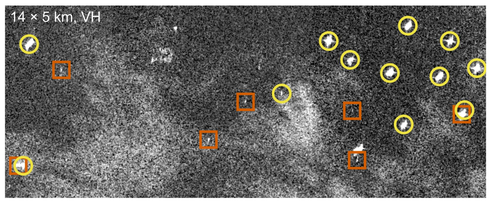
Label-Efficient Dark-Vessel Detection in SAR: Does SAR-Domain Pretraining Outperform Optical and ImageNet Transfer Across ViT and CNN?

John Roth **Kyle Wagner** **Liv d'Aliberti**

Johns Hopkins University

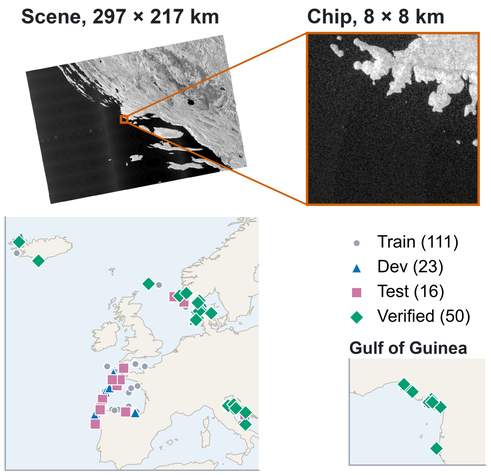
Answer: no. SAR pretraining beats optical in 6 of 16 comparisons and ImageNet in 3 of 16 (ImageNet wins 7); its clearest edge is the CNN with 12–28 labeled scenes.

Motivation



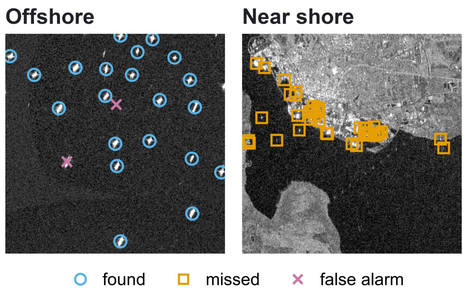
Dark vessels send no AIS, yet radar [2] still sees them. Human-checked labels are scarce: which pretraining stretches them?

Data: xView3-SAR [1]



- **Machine-made labels** (AIS + automated detection) for train, dev and test; they miss vessels.
- **50 human-verified scenes**: 8,642 vessels.

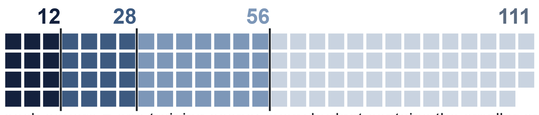
What the best detector finds



Best cell (ImageNet CNN, 111 scenes), 8 × 8 km verified crops. Verified F1: 0.724 offshore, 0.026 within 2 km of shore.

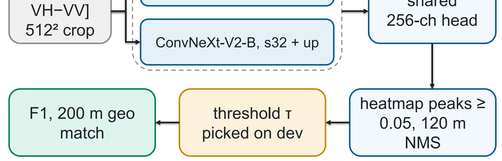
8 arms × 4 budgets = 32 cells

Role	ViT-B/16 [3]	ConvNeXt V2 [4]
Random	random init	random init
Optical	SatDINO [6] (fMoW-RGB)	BigEarthNet-S2 [8]
SAR	SARMAE [7] (SAR-1M)	BigEarthNet-S1 [8]
ImageNet	AugReg [5]	FCMAE [4]



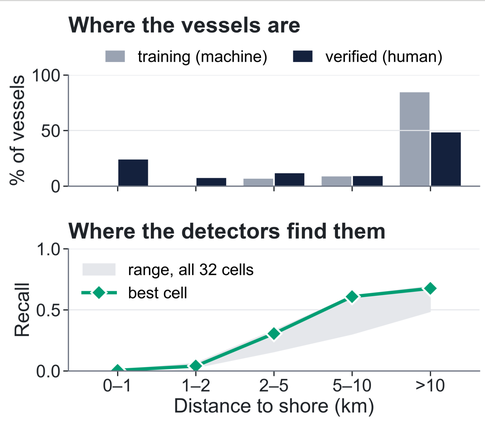
each square = one training scene; every budget contains the smaller ones

One shared point detector



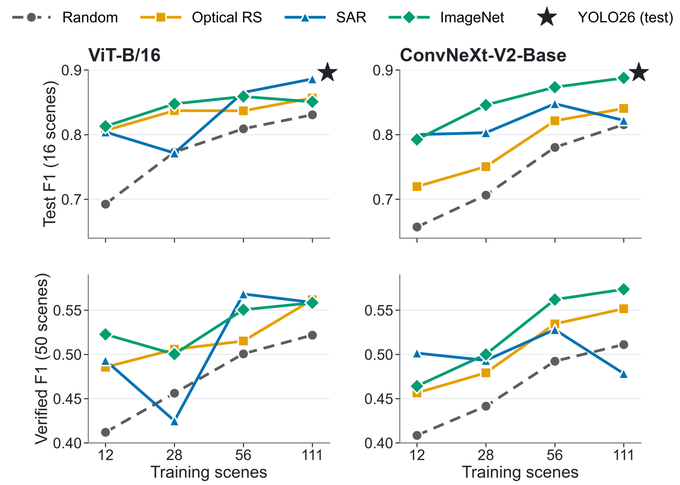
- CenterNet-style heatmap [9], focal loss [10]; one recipe and seed for every cell.
- 8 dev scenes pick checkpoint and threshold; test and verified scenes are scored **once**.

Recall falls toward the coast



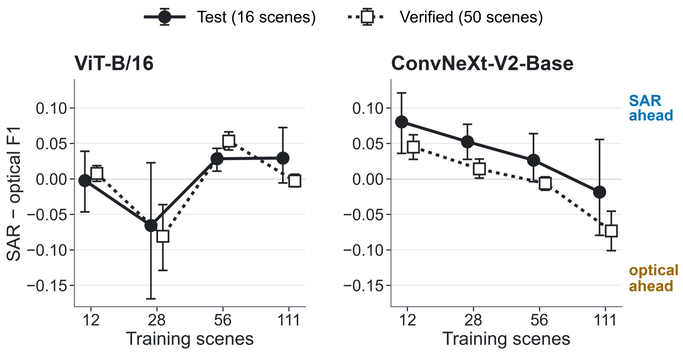
50 verified scenes; the band spans all 32 cells. Best-cell recall: 0.676 beyond 10 km, 0.005 within 1 km.

Pretraining helps most when labels are scarce



F1 with 200 m matching, each cell scored once at the threshold it picked on 8 dev scenes. Top: 16 test scenes; bottom: 50 verified scenes. ★ YOLO26 at 111 scenes, test only (0.896). Zero-shot LocateAnything-3B: 0.122 F1 on dev chips.

SAR vs. optical: no stable winner



How to read: each point is SAR F1 minus optical F1; above 0, SAR wins, and a bar clear of 0 is a win. Solid: test; dotted: verified. Bars: 95% intervals over 10,000 scene resamples; retraining moved test F1 by up to 0.043.

References

[1] F. S. Paolo et al. xView3-SAR: Detecting dark fishing activity using SAR imagery. NeurIPS 2022.

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[4] S. Woo et al. ConvNeXt V2: Co-designing and scaling ConvNets with masked autoencoders. CVPR 2023.

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[6] J. Straka and I. Gruber. SatDINO: A deep dive into self-supervised pretraining for remote sensing. arXiv:2508.21402, 2025.

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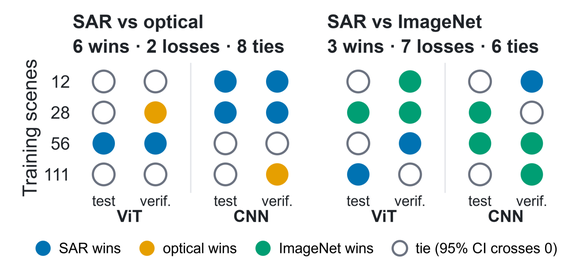
[8] K. N. Clasen et al. reBEN: Refined BigEarthNet dataset. IGARSS 2025.

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[10] T.-Y. Lin et al. Focal loss for dense object detection. ICCV 2017.

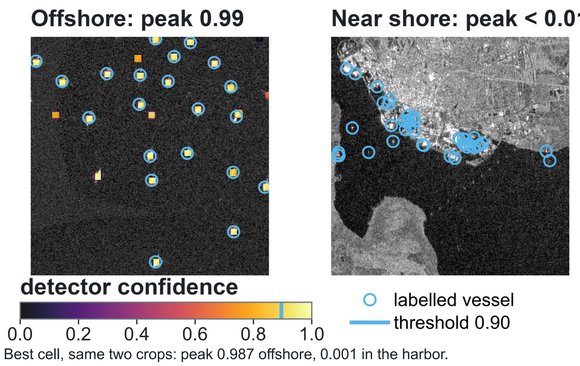
Findings

1. **Pretraining pays off with few labels:** +0.063 to +0.143 test F1 at 12 scenes.
2. **SAR pretraining rarely wins:**



3. **The input adapter matters:** a seeded 3-channel stem lifts BigEarthNet-S2 above random at every budget.
4. **No detector works near shore:** recall ≤ 0.021 within 2 km in every cell.

Inside the detector



Best cell, same two crops: peak 0.987 offshore, 0.001 in the harbor.

Limits

- One seed per cell; 8 dev scenes for selection.
- Machine-made test labels; in-region test scenes.
- Checkpoints also differ in objective and data.
- In-region test: Sentinel-1 revisits put some test scenes near training scenes, so test F1 measures in-region generalization, not generalization to new regions.

Open questions & next steps

- Would joint SAR + optical foundation models win?
- Does SAR trail ImageNet with matched pretraining?
- **Next: make detectors work near shore** (recall ≤ 0.021 within 2 km): train on human-labeled near-shore vessels, add a distance-to-shore input, and set shore-aware thresholds.